

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon IAD120, VA -- station KIAD, America/New_York
Committed pair OSM 693390022 -> 990537857
Amazon IAD120 / Amazon IAD122
Facade gap 178.5 m between the two halls
Weather record 43,763 real hourly records from KIAD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3805 C at 345 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-03-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.521	18.0	4.139	1.239
01	mechanical	yes	switch budget	4.521	18.0	4.139	1.179
02	mechanical	yes	switch budget	4.241	18.0	3.029	1.348
03	mechanical	yes	switch budget	3.518	18.0	1.266	1.248
04	mechanical	yes	switch budget	3.136	18.0	0.809	1.177
05	mechanical	yes	switch budget	2.551	18.0	0.249	1.353
06	mechanical	yes	switch budget	1.191	18.0	-0.861	1.168
07	mechanical	yes	switch budget	2.856	18.0	0.249	1.822
08	mechanical	yes	switch budget	6.333	18.0	5.560	2.014
09	mechanical	yes	switch budget	10.186	18.0	11.468	2.543

10	mechanical	yes	switch budget	11.646	18.0	13.890	2.323
11	mechanical	yes	switch budget	15.443	18.0	17.780	1.767
12	mechanical	no	dry-bulb	18.306	18.0	19.440	1.423
13	mechanical	no	dry-bulb	19.896	18.0	21.110	1.164
14	mechanical	no	dry-bulb	22.433	18.0	22.050	1.478
15	mechanical	no	dry-bulb	22.677	18.0	22.780	1.119
16	mechanical	no	dry-bulb	23.057	18.0	22.780	1.231
17	mechanical	no	dry-bulb	22.688	18.0	22.220	1.411
18	mechanical	no	dry-bulb	22.189	18.0	21.110	1.381
19	mechanical	no	dry-bulb	20.764	18.0	18.890	1.428
20	mechanical	no	dry-bulb	20.086	18.0	17.780	1.718
21	mechanical	no	dry-bulb	18.603	18.0	15.560	1.794
22	FREE	yes	-	15.765	18.0	12.220	1.399
23	FREE	yes	-	15.733	18.0	13.330	1.131

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.521 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.521 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.241 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.518 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.136 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.551 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.191 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.856 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.333 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.186 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.646 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.443 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.306 C against a 18.0 C limit, so it fails by 0.306 C. It would take a limit 0.306 C higher, or a bound 0.306 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.896 C against a 18.0 C limit, so it fails by 1.896 C. It would take a limit 1.896 C higher, or a bound 1.896 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.433 C against a 18.0 C limit, so it

fails by 4.433 C. It would take a limit 4.433 C higher, or a bound 4.433 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.677 C against a 18.0 C limit, so it fails by 4.677 C. It would take a limit 4.677 C higher, or a bound 4.677 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.057 C against a 18.0 C limit, so it fails by 5.057 C. It would take a limit 5.057 C higher, or a bound 5.057 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.688 C against a 18.0 C limit, so it fails by 4.688 C. It would take a limit 4.688 C higher, or a bound 4.688 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.189 C against a 18.0 C limit, so it fails by 4.189 C. It would take a limit 4.189 C higher, or a bound 4.189 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.764 C against a 18.0 C limit, so it fails by 2.764 C. It would take a limit 2.764 C higher, or a bound 2.764 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.086 C against a 18.0 C limit, so it fails by 2.086 C. It would take a limit 2.086 C higher, or a bound 2.086 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.603 C against a 18.0 C limit, so it fails by 0.603 C. It would take a limit 0.603 C higher, or a bound 0.603 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.765 C, which is 2.235 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.399 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.2104 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.733 C, which is 2.267 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.131 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.1778 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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